

MLCs might be good enough...

As we saw earlier in the article, single-level cells were traditionally considered to be more reliable than multi-level cells, a trend that current 3D NAND flash based technologies are attempting to change. The scores in favour of MLCs are up again, thanks to a finding by a team of scientists from the University of Toronto in collaboration with the Platforms Storage Team at Google. After analysing data from a number of custom-made drives over a six-year period, the researchers showed that while MLC based drives clearly had a larger number of raw bit error rates (RBERs) versus SLC drives, these were just as reliable overall. According to the report, "SLC drives, which are targeted at the enterprise market and considered to be higher-end, are not more reliable than the lower-end MLC drives."

which it reverts to idle power.

NVMe SSDs will also reduce the footprint and power consumption of large data centres, because on an average 10 SATA SSDs can be replaced with one NVMe SSD.

In the big league

Of late, we are seeing many key players launching all-flash solutions for large data centres. Trend-watchers note that enterprise and cloud players are beginning to consider flash based solutions not only for hot storage but for cold storage as well! No wonder 2016 is seeing a lot of activity on this front.

EMC recently launched the VMAX All Flash, a solution for mission-critical enterprise data centre applications, which it claims is more cost-effective than traditional methods. The all-flash storage array natively supports block, file, open systems and mainframe. VMAX All Flash includes innovative technologies such as Write Folding to improve flash durability while simultaneously delivering peak performance.

In March, Seagate demonstrated a new PCIe flash drive, which meets the Open Computer Project (OCP) specifications and delivers throughput of 10 gigabytes-per-second (GB/s). The add-in card bundles multiple small-sized, energy-efficient, consumer-grade M.2 SSDs. It accommodates 16-lane PCIe slots and supports NVMe. Seagate's new solution includes capabilities like bifurcation of PCIe lanes at boot-up, out-of-band temperature and performance measurements, and the management of airflow and fan control in out-of-band fashion, in order

to comply with OCP specifications. OCP, an initiative driven mainly by Facebook, is a collaborative effort to redesign hardware to meet the increasing demands on IT infrastructure, especially in large data centres with thousands of servers.

All-new tech, considered as new category

Although Intel Corporation and Micron Technology have showcased some 3D NAND designs, it is clear that they are betting on something bigger for the future. They go so far as to call their 3D XPoint technology the first new memory category since the introduction of NAND flash in 1989!

Demonstrated last year, 3D XPoint technology is a non-volatile memory that can provide extremely fast access to large sets of data. It is thousand times faster and thousand times more durable than NAND, and ten times denser than conventional memory. According to an Intel press release, "An innovative, transistor-less cross-point architecture creates a three-dimensional checkerboard where memory cells sit at the intersection of word lines and bit lines, allowing the cells to be addressed individually. As a result, data can be written and read in small sizes, leading to faster and more efficient read/write processes."

Picture 128 billion densely-packed memory cells, connected with perpendicular conductors. These cells store a single bit of data each, and can be accessed and written or read by varying the amount of voltage sent to each selector. The storage parts and the selector will be made of special chalcogenide materials.

Unlike NAND, which works by trapping electrons in the cell, here writing to the drive will fundamentally change the actual material of each memory cell. This new physical concept, along with the transistor-free cross-point architecture, gives Intel's 3D XPoint tech an enviable combination of high density, capacity and performance at a low cost. While current versions of the technology can store 128 gigabits per die across two stacked memory layers, in the future it is possible to increase the number of memory layers as well as increase die capacity using traditional lithographic pitch scaling.

This year, the company might bring to the market a new line of SSDs based on Intel Optane technology, which will combine 3D XPoint with an advanced system memory controller, interface hardware and software intellectual property. Intel dual-inline-memory-modules based on XPoint are also on the anvil for the data centre.

According to some expert reviews, XPoint chips can deliver more than 95,000 input/output operations per second (IOPS) at a 9-microsecond latency, compared to 13,400 IOPS and 73ms latency for flash in an NVMe based SSD. While that triggers awake our wildest dreams, there are also some constraints to think about, as Guy Blalock, co-CEO of IM Flash (the semiconductor company founded by Intel and Micron) pointed out at the Industry Strategy Symposium hosted by the SEMI trade group.

One possible problem with the new materials used in the design is their potential for cross-contamination. To mitigate this, they have to develop a lot of process steps for layer-over-layer deposition with a lot of diffusion and chemical vapour deposition. Plus, the 100+ new materials used by 3D XPoint also raise supply chain challenges. The design also requires more machines running process steps, cutting fab throughput by an estimated 15 per cent. This could increase capital expense and space needed by three to